

练习二十二 恒定电流

1. $v_A < v_B$ 、 $j_A < j_B$ 、 $E_A < E_B$

2. 相同、不同

3. (2); 4. (3)

5. (1) $R = \frac{L}{\sigma S} = \frac{2}{5.7 \times 10^7 \times 20 \times 80 \times 10^{-6}} = 2.2 \times 10^{-5} (\Omega)$

(2) $I = \frac{U}{R} = \frac{50 \times 10^{-3}}{2.2 \times 10^{-5}} = 2280 (A)$

(3) $j = \frac{I}{S} = \frac{2280}{20 \times 80 \times 10^{-6}} = 1.425 \times 10^6 (A \cdot m^{-2})$

6. (1) $I = \frac{\varepsilon_1 - \varepsilon_3}{R_1 + R_2 + R_5 + R_4 + r_1 + r_3} = \frac{12 - 8}{2 + 2 + 2 + 1 + 1 + 1} = \frac{4}{9} (A)$

$$U_{ab} = I(R_2 + R_5 + r_3) + \varepsilon_3 - \varepsilon_2 = \frac{4}{9} \times 5 + 8 - 10 = \frac{2}{9} (V)$$

(2) $I_1 = I_2 + I_3$

$$I_1(R_1 + r_1 + R_4) + I_2(R_3 + r_2) + \varepsilon_2 - \varepsilon_1 = 0$$

$$-I_3(R_2 + r_3 + R_5) + I_2(R_3 + r_2) + \varepsilon_2 - \varepsilon_3 = 0$$

代入数据解得: $I_1 = \frac{13}{28} = 0.46 (A)$